

## ATIVIDADE DE APOIO – ENSINO MÉDIO

**Componente Curricular:** Fundamentos dos compostos de Carbono

**Data:** 20/03/2025

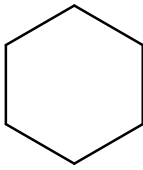
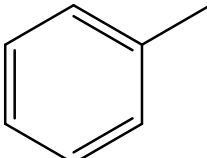
**Professor (a):** Gabriel H. M. Esteves

**Série:** 1ª EM

**Assunto (s):** Tarefa e Gabarito – Lista 3

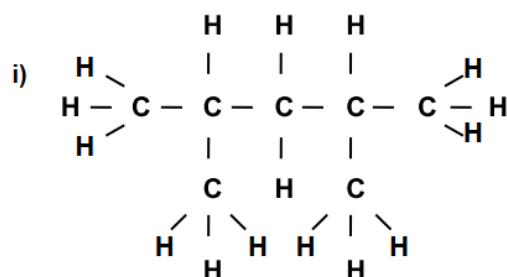
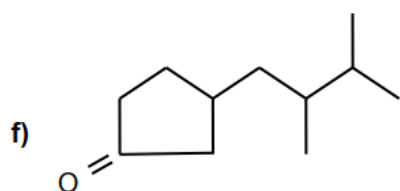
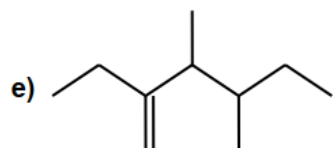
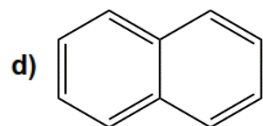
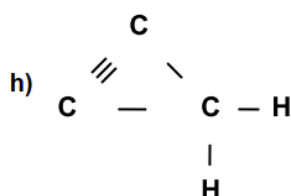
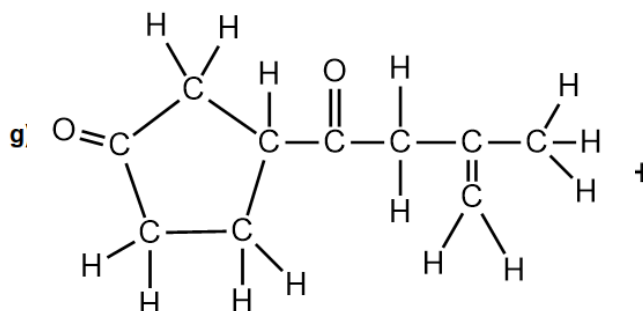
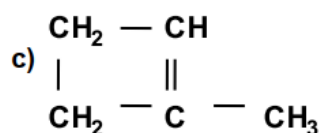
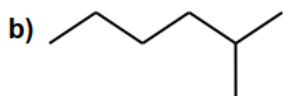
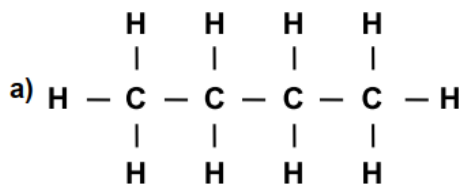
### Lista de exercícios 3

**01.** Complete os espaços indicados da Tabela abaixo com as fórmulas pedidas em cada coluna. Não é necessário colocar informações nas áreas hachuradas:

| Fórmula molecular | Fórmula estrutural simplificada                                                                                                                                               | Fórmula bastão                                                                        |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                   | $  \begin{array}{c}  \text{CH}_3 \\    \\  \text{H}_3\text{C}-\text{CH}-\text{CH}_3  \end{array}  $                                                                           |                                                                                       |
|                   |                                                                                                                                                                               |   |
|                   |                                                                                                                                                                               |  |
|                   | $  \begin{array}{c}  \text{CH}_3 \qquad \text{O} \\    \qquad \parallel \\  \text{H}_3\text{C}-\text{CH}-\text{CH}-\text{C}-\text{CH}_3 \\    \\  \text{CH}_3  \end{array}  $ |                                                                                       |
|                   |                                                                                                                                                                               |  |

**Obs:** Fórmulas bastão são um tipo de fórmula estrutural já que mostram como está a organização dos átomos da substância orgânica.

**02.** Em cada caso abaixo indique as outras formas de representar a cadeia carbônica (Fórmula estrutural simplificada, bastão e molecular):



**03.** Considerando a fórmula molecular  $C_3H_8O$ , escreva a fórmula bastão de três diferentes substâncias orgânicas sendo duas de cadeia homogênea e uma de cadeia heterogênea.

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**Exercícios complementares**

Faça os exercícios do Livro (Lumen, volume 4)

Pág. 115 – todos

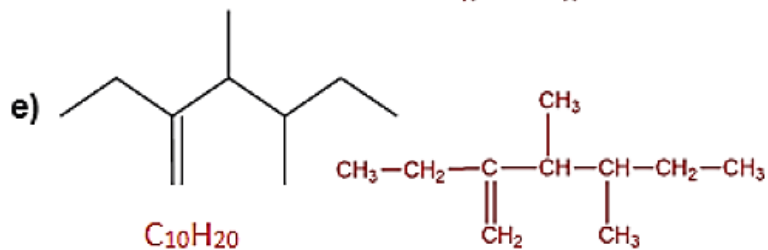
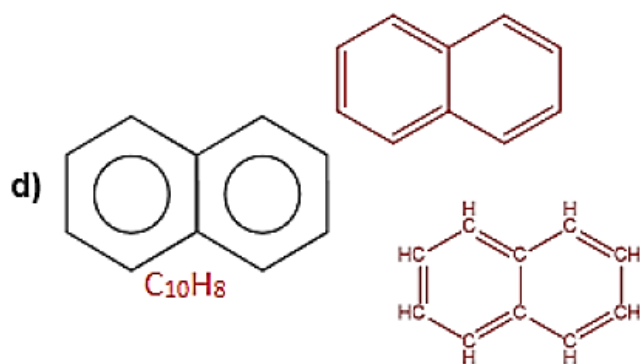
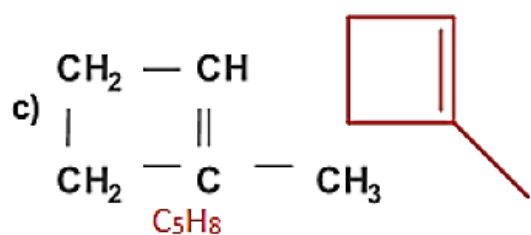
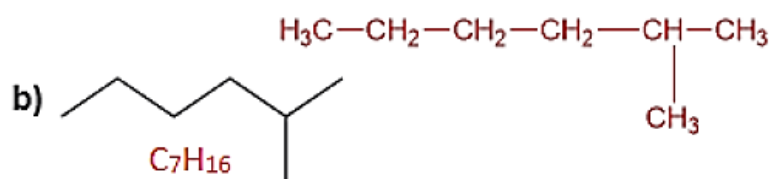
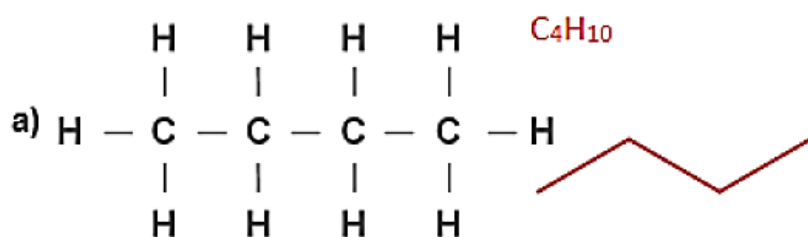
Pág. 116 – 1, 2, 3, 5 só o item (b), Superação só o item (a)

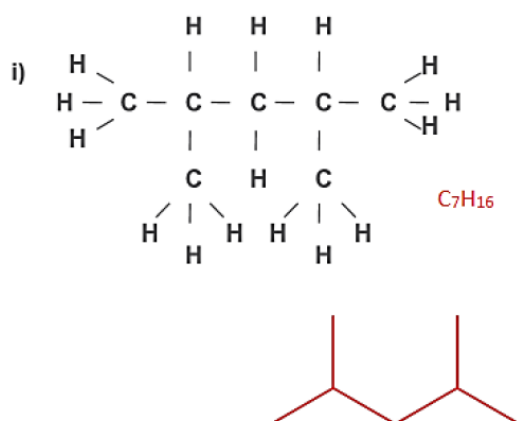
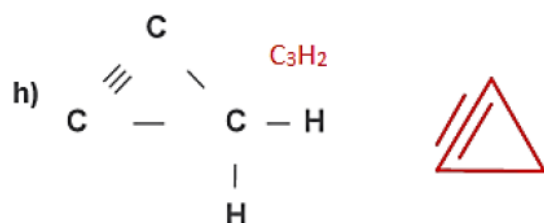
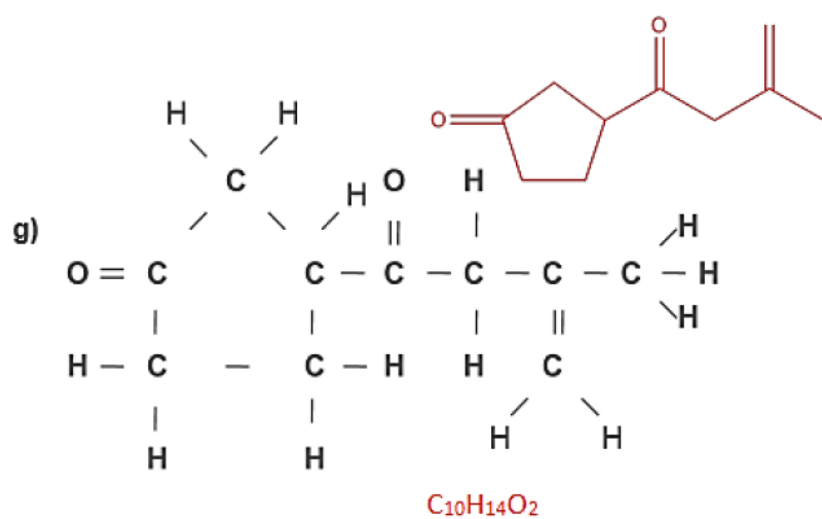
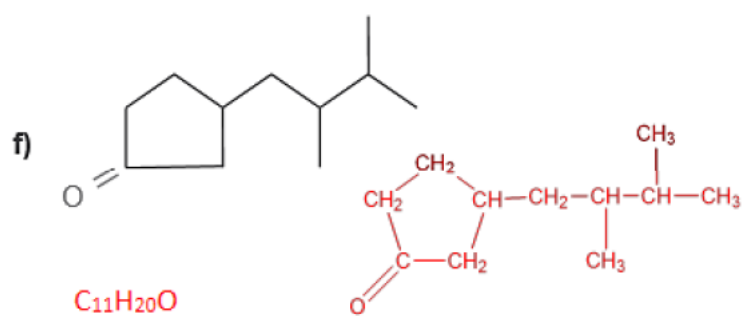
## GABARITO E RESOLUÇÕES

01.

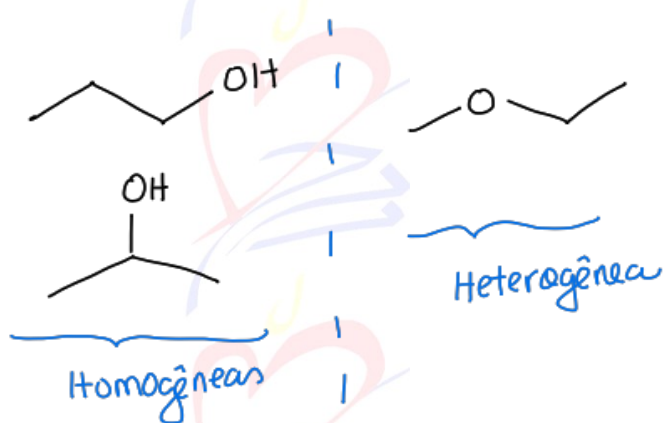
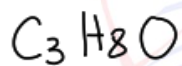
| Fórmula molecular                   | Fórmula estrutural simplificada                                                                                                                                                                                                                          | Fórmula bastão |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>C<sub>4</sub>H<sub>10</sub></b>  | $\begin{array}{c} \text{CH}_3 \\   \\ \text{H}_3\text{C}-\text{CH}-\text{CH}_3 \end{array}$                                                                                                                                                              |                |
| <b>C<sub>6</sub>H<sub>12</sub></b>  | $\begin{array}{c} \text{H}_2 \\   \\ \text{H}_2\text{C}-\text{C}-\text{CH}_2 \\   \quad   \\ \text{H}_2\text{C}-\text{C}-\text{CH}_2 \\   \\ \text{H}_2 \end{array}$                                                                                     |                |
| <b>C<sub>6</sub>H<sub>14</sub></b>  | $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$                                                                                                                                                                                |                |
| <b>C<sub>7</sub>H<sub>14</sub>O</b> | $\begin{array}{c} \text{CH}_3 \quad \quad \text{O} \\   \quad \quad \parallel \\ \text{H}_3\text{C}-\text{CH}-\text{CH}-\text{C}-\text{CH}_3 \\   \\ \text{CH}_3 \end{array}$                                                                            |                |
| <b>C<sub>7</sub>H<sub>8</sub></b>   | $\begin{array}{c} \text{H} \quad \quad \text{CH}_3 \\ \diagup \quad \diagdown \\ \text{HC} \quad \quad \text{C} \\ \parallel \quad \parallel \\ \text{HC} \quad \quad \text{CH} \\ \diagdown \quad \diagup \\ \text{C} \quad \quad \text{H} \end{array}$ |                |

02.





③

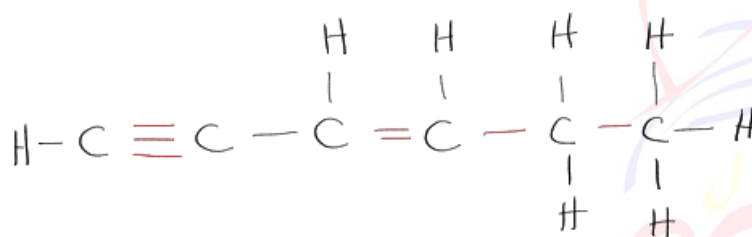


Complementares – Gabaritos e Resoluções

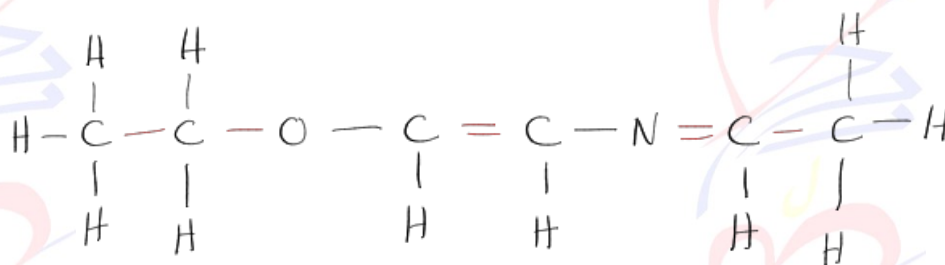
Lumen vol.4 (Pág 115 as)

①

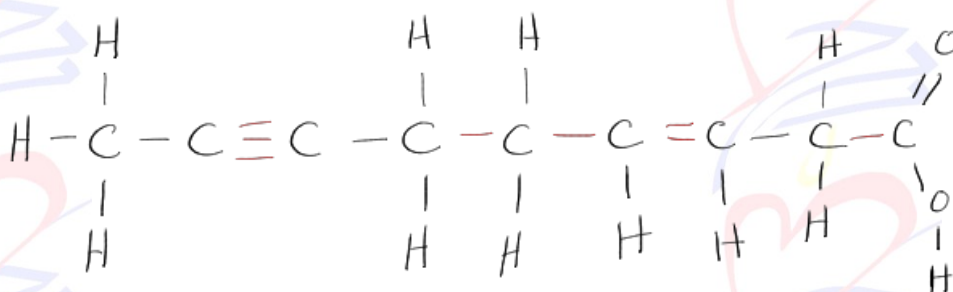
a)



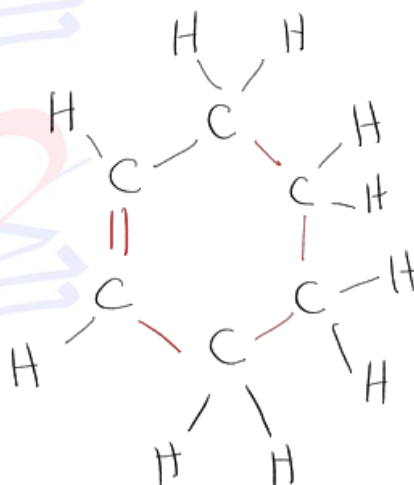
b)



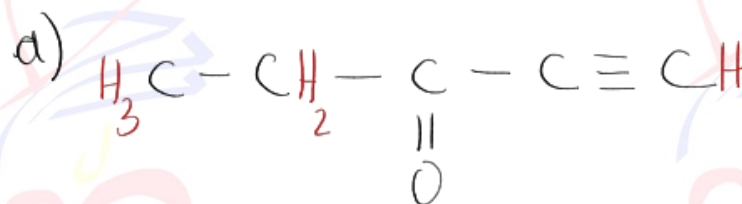
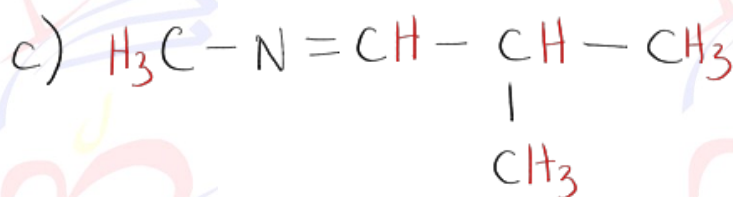
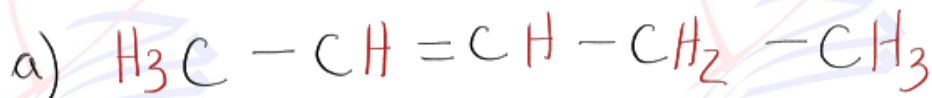
c)



d)

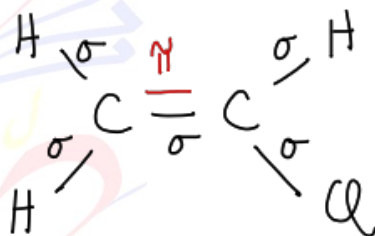


②

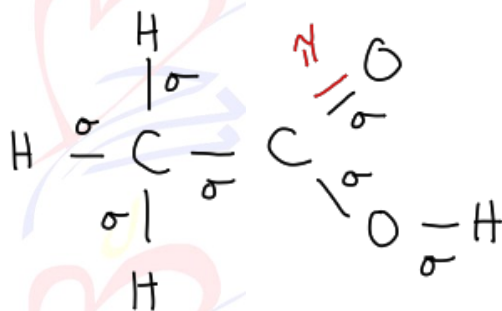


③

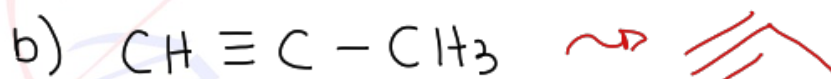
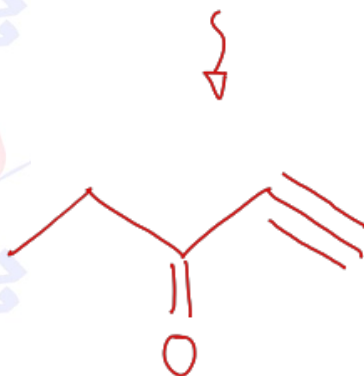
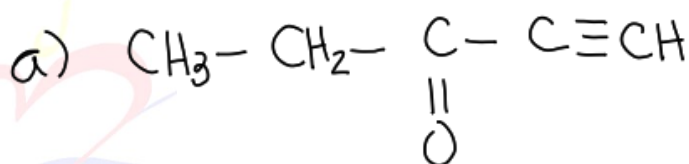
a) No C-1 e no C-2 a hibridização é de tipo  $sp^2$ .



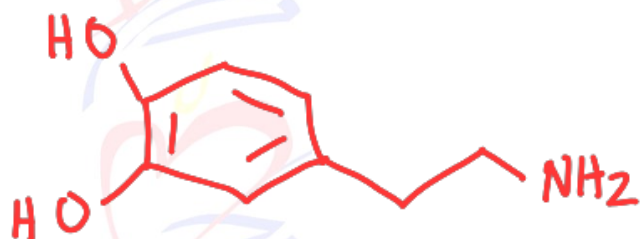
b) No C-1 a hibridização é  $sp^3$  e  
no C-2 a hibridização é  $sp^2$



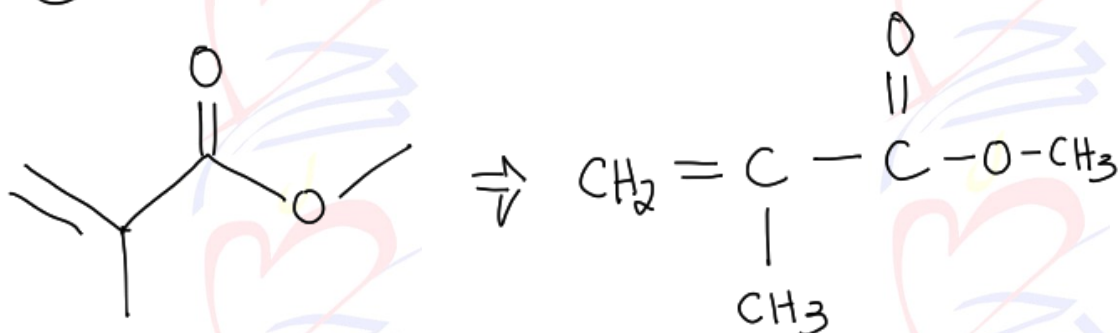
④



⑤



⑥ Alternativa [E]

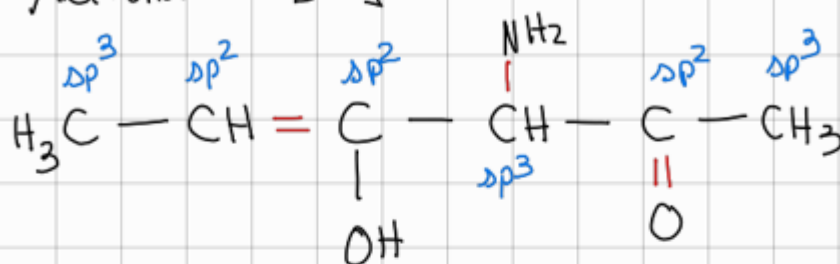


fórmula molecular =  $C_5H_8O_2$

14  $\sigma$  e 2  $\pi$

Consolidando saberes pág 116

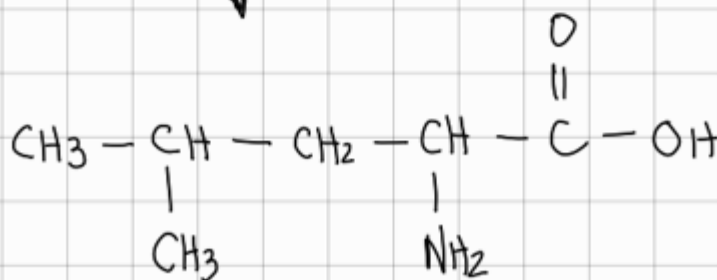
① Alternativa [C]



② Alternativa [B]



fórmula molecular



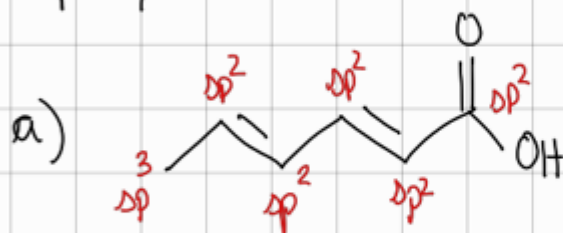
④ Não fazer !

⑤ a) Não fazer !

b)

| Carbono | hibridização |
|---------|--------------|
| 1       | $sp^3$       |
| 2       | $sp^2$       |
| 3       | $sp^3$       |
| 4       | $sp^3$       |

Superação



b) Não fazer !

